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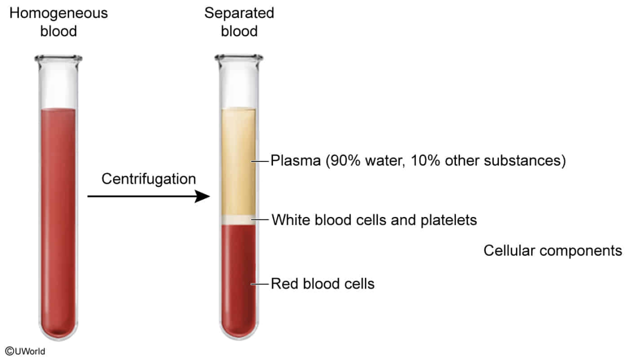
Transferrin circulating in the blood consists of 679 amino acids, but it is composed of 698 amino acids when it is post-translationally extracted from liver cells, the primary site of transferrin synthesis. Given this information, researchers would most likely be able to extract and isolate the shorter transferrin peptide from:

- A. lymph. [9%]
- ✔ B. the liquid portion of blood. [53%]
- C. all cellular elements of blood. [18%]
- D. the nucleus of a liver cell. [19%]

Correct
53% answered correctly

Explanation:



Blood is the only fluid connective tissue in the body and is composed of living blood cells and nonliving plasma. The cellular components of blood originate from the bone marrow: Erythrocytes, which have a role in oxygen transport; leukocytes, which contribute to immune defense; and platelets, which are crucial in blood clotting. Plasma is the **liquid portion of blood** and is approximately 90% water, with the remaining 10% composed of other substances (eg, proteins, respiratory gases).

As stated in the question, transferrin isolated from liver cells has 698 amino acids; however, circulating transferrin has 679 amino acids. The 19 missing amino acids likely make up a signal sequence, a short amino acid sequence present at the N terminus of immature proteins. The signal sequence directs proteins to the rough endoplasmic reticulum for entry into the secretory pathway before being cleaved from the polypeptide. In this case, transferrin is trafficked through the entire pathway for secretion into extracellular space. Once secreted from liver cells, the **shortened transferrin** enters circulation and is **incorporated into** the **plasma**.

(Choice A) The passage states that transferrin binds free iron in blood, *not* in lymph. Hydrostatic pressure (the force exerted by blood against vessel walls) forces some liquid and smaller molecules to leak out of the capillaries. At this point, the liquid is referred to as *interstitial fluid*, and it is present in the space between the blood vessels and surrounding cells. The interstitial fluid is then collected by the lymphatic vessels. The fluid, now called lymph, is filtered through lymph nodes (pockets on the lymphatic vessels containing white blood cells) and returned to the circulatory system.

(Choice C) Transferrin functions to shuttle free iron from *blood plasma* into developing erythrocytes. Although lingering transferrin may be present in mature erythrocytes, it is not present in *all* blood cells (ie, leukocytes).

(Choice D) DNA replication and RNA transcription occur in the nucleus. Translation occurs in the cytoplasm. Therefore, the transferrin peptide can be isolated from the cytoplasm of a liver cell but not from the nucleus.

Educational objective:

Plasma is the liquid portion of blood. It is made up of water, electrolytes, gases, hormones, nutrients, metabolic waste, and proteins. Although derived from blood, lymph is not a component of blood.

Time spent:0 sec
Subject:
Biology

QID:401699
System:
Circulation and Respiration

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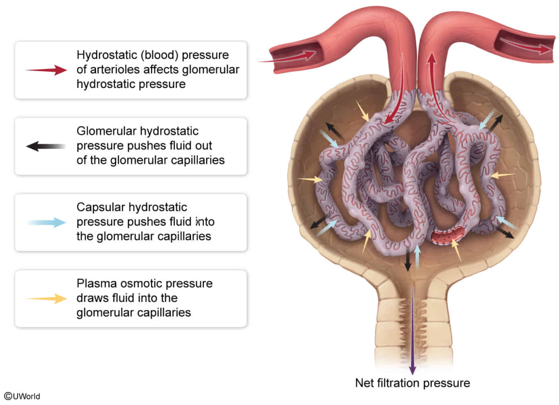
Iron buildup in the liver can lead to scarring and loss of liver function. If a change in holoTf excretion rate were to cause this form of liver damage in a patient receiving blood transfusions, the rate at which holoTf is filtered from the glomerular capillaries in this patient would most likely be:

- A. decreased due to increased Na^+ reabsorption in the nephron. [6%]
- ✔ B. decreased due to decreased hydrostatic pressure within the glomerulus. [57%]
- C. increased due to increased urinary output from the kidney. [12%]
- D. increased due to increased hydrostatic pressure within the glomerulus. [23%]

Correct
57% answered correctly

Explanation:

Glomerular filtration



The kidneys consist of functional units known as nephrons that filter waste products (eg, excess iron) out of the blood while reabsorbing useful solutes back into the blood.

Blood initially enters the kidneys through the renal arteries, which separate into smaller, afferent arterioles that carry blood toward the nephrons. Afferent arterioles then branch into a ball-like network of capillaries known as the glomerulus. The glomerulus itself is surrounded by a nephron component called the Bowman's capsule that collects leaked glomerular fluid (ie, renal filtrate). The blood then exits the glomerulus through the efferent arterioles. Efferent arterioles later branch into a second series of capillaries that encapsulate the remaining portions of the nephron, facilitating the movement of components out of the filtrate and into the blood, or vice versa.

The **glomerular filtration rate (GFR)** is the rate (amount per unit time) at which fluid is filtered out of the glomerular capillaries (collectively known as the glomerulus) and into the **Bowman capsule**. GFR is proportional to the hydrostatic (blood) pressure within the glomerulus. As such, GFR can be expected to decrease if the afferent arterioles constrict (ie, because less blood enters the glomerulus) or the efferent arterioles dilate (ie, because more blood exits the glomerulus).

In this scenario, a patient receiving blood transfusions exhibits liver damage caused by iron *buildup*. The passage states that iron-bound transferrin (holoTf) could be filtered through the kidney. Therefore, if **GFR decreased** in the kidney, the likely result would be **increased retention of iron-bound transferrin** (holoTf) in the blood, which would contribute to **iron buildup** in the liver. **Decreased hydrostatic pressure** within the glomerulus can lead to *decreased* GFR (and consequently decreased holoTf filtration rates) because the flow rate of filtered fluid through the kidney will decrease.

(Choice A) Increased Na^+ reabsorption in the nephron, which is stimulated by the adrenal hormone aldosterone, also leads to increased water reabsorption. This results in increased blood pressure without altering blood osmolarity (blood solute concentration). Increased blood pressure increases GFR, which would likely decrease (*not* increase) holoTf retention.

(Choices C and D) Increased hydrostatic pressure within the glomerulus would likely cause *increased* GFR, leading to *increased* urinary output. This would likely *increase* the filtration rate of holoTf from the blood, *decreasing* (*not* increasing) iron retention in the body.

Educational objective:
Glomerular filtration rate is a measure of the rate at which fluid is filtered into the Bowman

capsule from the glomerulus.

Time spent: 0 sec	**QID**:401700
Subject: Biology	**System:** Circulation and Respiration

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In Hbb^{th-3}/Hbb^{th-3} mice that received apoTf injections, which of the following tables best supports the conclusion that holoTf excretion is responsible for decreased non–transferrin-bound iron in the blood (NTBI)?

A.

Days post-injection	Blood NTBI (ng/mL)	Urinary holoTf (μg/mL)
0	250	1.0
15	50	1.0

[4%]

B.

Days post-injection	Blood NTBI (ng/mL)	Urinary holoTf (μg/mL)
0	50	0.5
15	250	1.0

[6%]

✔ C.

Days post-injection	Blood NTBI (ng/mL)	Urinary holoTf (μg/mL)
0	250	0.5
15	50	1.0

[87%]

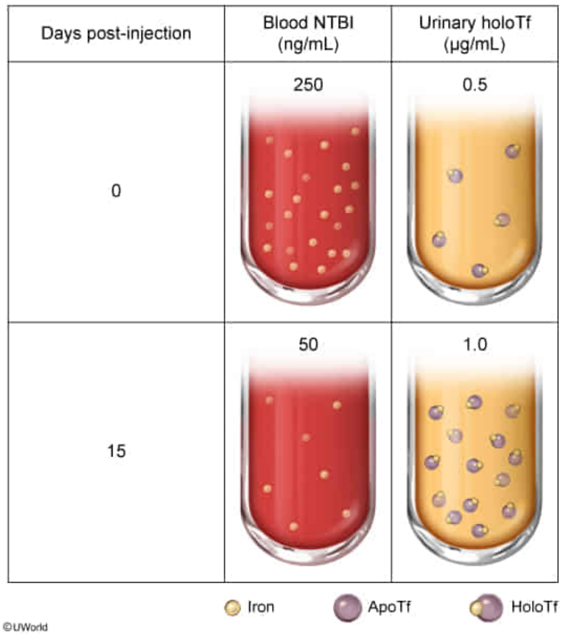
D.

Days post-injection	Blood NTBI (ng/mL)	Urinary holoTf (μg/mL)
0	50	0.5
15	250	0.5

Correct

87% answered correctly

Explanation:



In the passage, transferrin is defined as a blood protein that binds free iron in circulation. When transferrin is bound to iron, the iron-transferrin complex is referred to as holoTf, but when not bound to iron, the lone transferrin molecule is referred to as apoTf. Patients with beta-thalassemia require regular blood transfusions, resulting in excess blood iron buildup that can have detrimental effects. Experimenters predicted that injected apoTf could bind the excess free iron in the blood and prevent it from depositing in the organs.

To test this prediction, researchers administered apoTf injections to *Hbb*$^{th-3}$/*Hbb*$^{th-3}$ mice (a beta-thalassemia animal model) that had been given blood transfusions. The researchers found that non–transferrin-bound iron in the blood (NTBI) *decreased* following apoTf administration.

Based on these results, the researchers hypothesized that NTBI decreased because the **administered apoTf** was able to **bind the excess iron**, forming holoTf and leading to increased iron excretion rather than iron deposition in the organs. Because increased iron excretion can be detected as an increase in *urinary* holoTf, this hypothesis is best supported by the data in **Choice C**, which is the **only table that shows *both* a decrease in NTBI** (from 250 to 50 ng/mL) *and* an **increase** in **urinary holoTf** (from 0.5 to 1.0 μg/mL).

(Choices A and D) In these tables, no changes in urinary holoTf levels are documented between days 0 and 15 after apoTf injection. These findings do *not* correspond to the predicted increase in urinary holoTf level.

(Choice B) Although this table shows an increase in urinary holoTf level following apoTf injection, there is also an *unexplained* increase in blood NTBI level. Because more iron is leaving the body through the urine, blood iron levels are expected to be *lower* on day 15 than on day 0.

Educational objective:

In the case of experiments that involve assessment of a dependent variable before and after administration of a particular treatment, a proposed hypothesis can be supported or not supported by assessing how the dependent variable changes before and after treatment administration.

Time spent:0 sec	**QID**:401701
Subject: Biology	**System:** Circulation and Respiration

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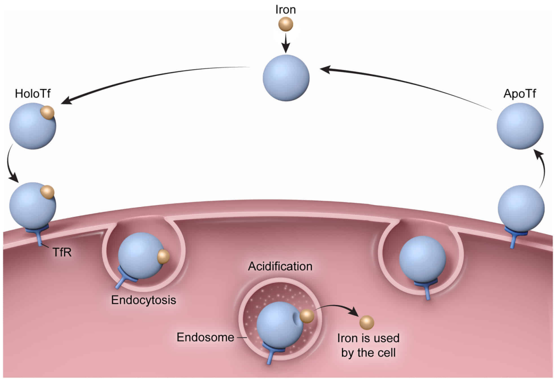
In cells expressing TfR, decreased endocytic vesicle pH during normal iron metabolism would trigger the recycling of:

- A. only holoTf to the extracellular space. [14%]
- B. only apoTf to the extracellular space. [12%]
- C. apoTf to the extracellular space and TfR to the cell membrane. [44%]
- D. holoTf to the extracellular space and TfR to the cell membrane. [28%]

Correct
44% answered correctly

Explanation:

Recycling of ApoTf and TfR



Endocytosis is a type of transport in which molecules that are otherwise unable to cross the plasma membrane enter a cell through vesicles (endosomes) formed by the plasma membrane. Endocytosis can take place in one of three ways: pinocytosis, phagocytosis, or receptor-mediated endocytosis.

Receptor-mediated endocytosis is a highly specific process that begins when a particular extracellular molecule (ligand) binds to its corresponding receptor on the cell surface. This binding triggers the invagination of the plasma membrane, which then closes around the bound extracellular content and pinches off from the rest of the membrane to become a vesicle inside the cell. The extracellular material remains trapped in the vesicle, which matures into an endosome.

The passage states that iron-bound transferrin is referred to as holotransferrin (holoTf), and non−iron-bound transferrin is known as apotransferrin (apoTf). HoloTf first binds the cell surface transferrin receptor (TfR), forming the holoTf-TfR complex and triggering vesicle formation. According to the passage, **acidification** (ie, decreased pH) **of the endosome** results in the release of iron from holoTf, which then becomes apoTf. In erythrocyte precursor cells, that iron is then incorporated into heme groups. The remaining complex components, **apoTf and TfR**, are **recycled to the membrane** to facilitate the uptake of another iron molecule.

(Choices A and D) HoloTf is the transferrin-iron complex. When iron is released from the complex, the remaining portion of the complex that is recycled to extracellular space is apoTf, *not* holoTf. The recycling of apoTf allows the molecule to bind and shuttle another iron molecule from the blood plasma into a developing erythrocyte.

(Choice B) TfR must also be recycled to the cell surface to bring more holoTf into the cell.

Educational objective:
Endocytosis is a mechanism by which extracellular contents are transported into the cell within a vesicle formed from the plasma membrane. During receptor-mediated endocytosis, specific extracellular targets bind their corresponding receptors and are incorporated into intracellular vesicles that mature into endosomes.

Time spent: 0 sec
Subject: Biology

QID: 401702
System: Circulation and Respiration

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Individuals with beta-thalassemia are at risk for developing an enlarged spleen. The resulting impairment of spleen function in these individuals would NOT affect:

- A. immune responses against viral or bacterial infection. [7%]
- ✔ B. release of hormones regulating blood glucose concentration. [82%]
- C. removal of aged red blood cells from the circulation. [3%]
- D. storage of red blood cells. [6%]

Correct
81% answered correctly

Explanation:

Spleen function
• Filters aged/damaged RBCs • Reservoir for blood • Immune response (B cell activation site, housing macrophages)

The spleen is the largest lymphoid organ in the body. It is situated in the upper left portion of the abdominal cavity, just above the stomach and below the diaphragm. The spleen performs the following functions:

- **Filters** senescent (worn-out, aged) or damaged **red blood cells** (RBCs) from the blood. In addition to destroying foreign material, macrophages in the spleen trap and phagocytize aged RBCs and recycle their parts **(Choice C)**. (Note: The liver is also involved in the removal and reuse of aged RBCs.)

- Serves as a **reservoir for blood**. This blood can be released in the event of a sudden drop in blood volume **(Choice D)**.

- Participates in **immune function** by housing macrophages that phagocytize and destroy foreign substances and by serving as a site of B-cell activation **(Choice A)**. Each B cell expresses a unique antibody that recognizes and binds a specific pathogen, preventing it from entering host cells and marking it for phagocytosis. An activated B cell (ie, one that has come in contact with its corresponding pathogen) rapidly proliferates to produce memory and effector B cells. Effector B cells (ie, plasma cells) produce antibodies against the current infection, and memory B cells are responsible for long-term defense against a similar infection in the future. Individuals who do not have a functional spleen cannot mount B-cell responses as efficiently, and their lifetime risk of infection is increased.

The organ primarily responsible for releasing hormones that regulate blood glucose levels is the pancreas. When blood glucose is high, pancreatic beta cells release the hormone insulin to promote cells to take up glucose, which is then either metabolized for energy or stored as glycogen in muscle and liver cells. When blood glucose is low, pancreatic alpha cells release the hormone glucagon to promote the production of glucose (via gluconeogenesis) and the release of glucose from glycogen stores. Therefore, impairment of spleen function would *not* likely affect this pancreatic function.

Educational objective:
The spleen is involved in immune function, blood storage, and the filtration and removal of senescent and damaged red blood cells from the blood.

Time spent: 0 sec	**QID:** 401703
Subject: Biology	**System:** Circulation and Respiration

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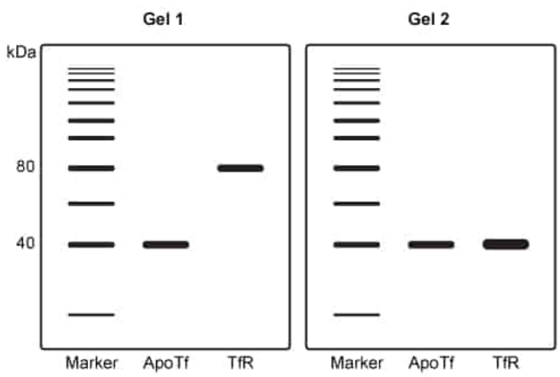
ApoTf and TfR were analyzed by sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) under nonreducing and reducing conditions. Based on the gels shown, which conclusion is LEAST likely to be true?

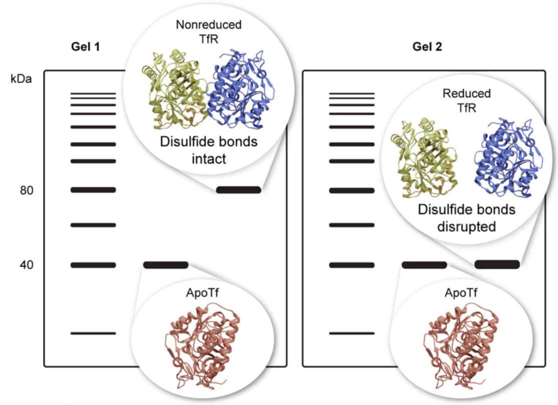


- A. Gel 1 represents the nonreducing SDS-PAGE gel. [21%]
- B. ApoTf has a smaller molecular weight than TfR. [15%]
- C. ApoTf and TfR were assigned a net negative charge. [10%]
- D. TfR is a monomeric protein lacking disulfide bridges. [52%]

Correct
51% answered correctly

Explanation:

SDS-PAGE under nonreducing & reducing conditions



SDS-PAGE = sodium dodecyl sulfate-polyacrylamide gel electrophoresis

Protein electrophoresis is a method of separating proteins according to factors that (depending on the specific technique used) can include shape, molecular weight, and charge. Sodium dodecyl sulfate (SDS) polyacrylamide gel electrophoresis (SDS-PAGE) is an electrophoresis technique used to separate proteins based on molecular weight (size). SDS is a strong anionic detergent that denatures and confers a net negative charge on the protein, eliminating its shape and charge as separation factors. When this technique is used, larger proteins migrate a shorter distance toward the positive electrode than smaller proteins.

However, SDS disrupts only noncovalent bonds. Therefore, a protein containing multiple covalently bound polypeptide subunits presents as a single band in SDS-PAGE electrophoresis. When SDS-PAGE is performed under reducing conditions, a reducing agent (eg, 2-mercaptoethanol) is added to disrupt disulfide bridges, allowing each subunit to separate as individual polypeptides.

In this question, analysis of apoTf and TfR by SDS-PAGE under nonreducing and reducing conditions is described. Because apoTf did not split into multiple bands in either gel and migrated to the same position (40 kDa) under both nonreducing and reducing conditions, it can be concluded that apoTf is a 40-kDa monomeric protein. Although TfR did not split into multiple bands, it did migrate to different positions within the gels depending on the experimental condition. In Gel 1, TfR appears as an 80-kDa band, and in Gel 2 it appears as a 40-kDa band. Accordingly, it can be concluded that when subjected to reducing conditions (ie, in Gel 2), the disulfide bridges holding TfR together are disrupted, and the 80-kDa TfR dimer splits into two 40-kDa subunits.

Based on this information, the following conclusions are likely to be **true**:

- Because Gel 1 contains the whole 80-kDa TfR protein, Gel 1 represents the *nonreducing* condition. In addition, this gel shows that apoTf has a *smaller* molecular weight (40 kDa) than TfR **(Choices A and B)**.

- Because of the nature of SDS application, both apoTf and TfR were assigned a net negative charge before undergoing electrophoresis **(Choice C)**.

Gel 2 shows that TfR *splits* into two 40-kDa subunits under reducing conditions; therefore, **TfR** is most likely a **dimeric**, *not* monomeric, protein.

Educational objective:
Comparing the results of reducing and nonreducing sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) (techniques that separate proteins based on molecular weight) can be useful in determining whether disulfide bonds contribute to the tertiary structure of a protein.

Time spent:0 sec	**QID**:401704
Subject:	**System:**
Biology	Circulation and Respiration